

Rebuilding Your Flood-Damaged Home

Testing and first-hand observation provide the best available information about how to rebuild.

by Robert Wendt and Heshmat Aglan

With more than 100,000 flood-damaged houses along the Gulf Coast facing replacement or reconstruction, homeowners and builders need access to the best available information on rebuilding. We have been conducting research for five years on the materials, systems, and methods that are best to use to repair flooded homes, making them more resistant to future flood damage, while simultaneously reducing energy costs and increasing home comfort. In addition, these methods have the potential to lower home flood insurance rates.

Our research focus has been on damage from wetting and drying in flooding where the water movement is primarily vertical. We have not studied, nor will we address, the impacts of storm surge or rapidly moving water that could cause structural damage. The following recommendations are based both on testing done by Tuskegee University and Oak Ridge National Laboratory and on observations taken from post-Katrina New Orleans.

Our studies have not yet evaluated the impact of chemical or biological contaminants—fuel oil or sewage, for example; therefore, some of our recommendations may not be appropriate for heavily contaminated environments. Floodwaters from unpolluted rivers or streams are likely to be uncontaminated. If you have concerns about possible contamination, contact EPA, which frequently tests floodwaters. We saw the same pollutants they identified in the New Orleans floodwater in the materials of the houses we visited. Some contaminants may leave visible clues; others don't. Heavy oils, for example, may be visible on the surface; lighter oils may be hard to see; while other chemicals are usually invis-



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Wood framing in this New Orleans house was undamaged by flooding.

ble. Some chemicals may leave a telltale smell. Testing is the only sure way to know, but it can be expensive.

FEMA Requirements

FEMA and the National Flood Insurance Program (NFIP) typically require that when substantial damage occurs to a home within a floodplain—with repairs costing at least 50% of the home's preflood market value—one of the following approaches must be employed.

Elevate. Raise the structure in its present location so that the lowest floor is at or above the base flood elevation (BFE). You must verify the BFE at your location, because it changes periodically.

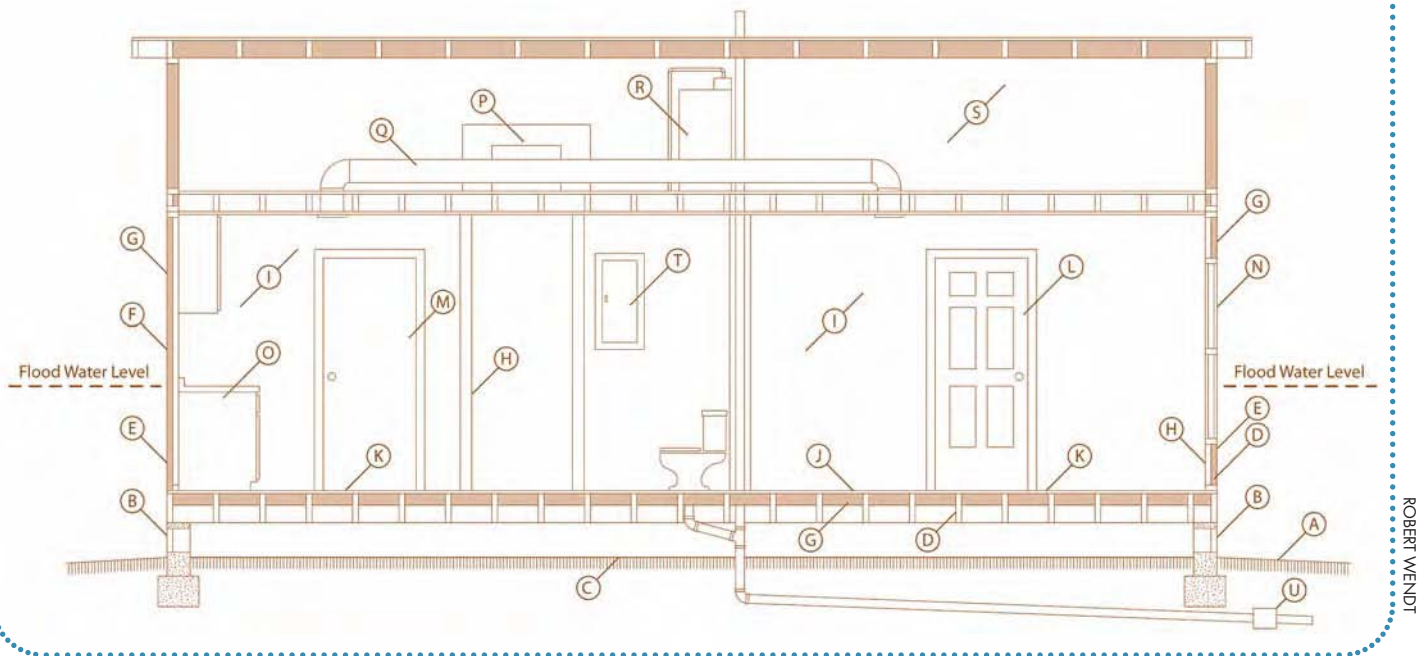
Relocate. Move the structure to a new, higher location, typically one outside the 500-year floodplain.

Acquire and demolish (buy out).

When the two previous options are impractical, the NFIP will buy out the homeowner and demolish the damaged home. The buyout will enable the homeowner to acquire another home in a new location.

When a house is not substantially damaged—when repairs cost less than 50% of the home's preflood market value—the reconstruction response could follow any of the above approaches (elevate, relocate, or acquire and demolish), or a wet floodproofing strategy could be used. Wet floodproofing allows for the probability that floodwater will enter the house and uses materials and systems that will sustain no more than cosmetic damage and will be restorable to preflood conditions at a reasonable cost. This approach is often accompanied by elevation or shielding of

Recommendations by Location



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Figure 1. Techniques for restoring each section of the house that is labeled with a letter are discussed in the article.

key utility equipment and contents to protect them from flooding. (A dry floodproofing approach that excludes floodwater within the house by sealing the walls and shielding openings against floodwater entry is not accepted for residential structures.) Improving the wet floodproofing characteristics of a home is essential to the restoration process.

Restoring the Building Envelope

Working in a flood-damaged home can be hazardous; preparing the home for reconstruction requires knowledge and training. Refer to the American Red Cross publication, *Repairing Your Flooded Home*, and to FEMA's recovery advisories for excellent guidance on a safe return to a flooded property and on prudent methods of recovery.

There are two key considerations when providing wet floodproofing: Either completely exclude floodwater from materials or encourage drying of materials that have become wet.

The "exclude water" approach is at best difficult to accomplish in typical residential construction. Some materials, such as plastic foam insulation and ceramic tile flooring, absorb little or no

water and effectively prevent water from entering that part of the wall or floor system. However, floodwater can find small cracks in those systems and bypass these impermeable materials, perhaps trapping water within other parts of the system. Therefore it is wise to provide for at least one means of escape for water in all systems exposed to flooding.

The "encourage drying" approach acknowledges the probability that water will get into a system and provides appropriate means of escape for that water. At least one means for water or water vapor to escape is identified and maintained and, where possible, drying in multiple directions is facilitated. The rate of drying needs to be consistent with the materials in the systems. Materials that are subject to rot or bacteriological growth—like wood—must be able to dry faster than materials—like concrete—that are not.

Testing has shown that while individual materials may perform satisfactorily, when they are combined into a typical housing assembly or system, their performance, or that of the materials around them, may degrade. It is therefore imperative that system performance be the criterion for judging flood damage resistance.

Reconstruction Techniques

Here are our location-by-location recommendations for restoring and rebuilding a home. All references cited in the text are to points shown in Figure 1.

Site drainage (A). Homes undergoing restoration may require regrading of the site around the house to promote drainage away from the structure and promote drying in the crawlspace. Provide a 5% (1 inch in 20 inches) slope away from the structure for a distance of 10 feet, if possible.

Foundation vents (B). Conventional foundation vents may become blocked by debris during flooding, and this blockage could induce unintended hydrostatic loads on the structure, damaging it. Operable flood vents that are closed before flooding starts and that open by themselves as the floodwater rises should replace conventional vents. (These vents are controlled by a float mechanism that allows them to close as the water recedes.) These vents have larger-than-normal openings, which minimizes the potential for blockage from debris. Replacement flood vents should be located as low on the wall as is practical, to promote draining of the crawlspace after a flood.

Crawlspace (C). Humidity in the crawlspace after flooding will remain high for an extended period of time. This can promote mold growth. All penetrations in the flooring above a crawlspace—such as pipes, wires, gaps, and so on—should be effectively sealed to prevent excess moisture and mold from entering the interior of the home.

If the crawlspace becomes accessible during reconstruction, consideration should be given to regrading the dirt floor to drain the crawlspace to the perimeter of the house. After the initial drying of the crawlspace floor has been accomplished, it is desirable to cover the dirt floor completely with a 6-mil polyethylene vapor barrier. This will help reduce future moisture levels in the crawlspace. While sealed crawlspaces are being successfully used as an energy efficiency strategy in many parts of the country, they cannot be used in potentially flood-prone areas. A sealed crawlspace could induce hydrostatic pressure on the structure, potentially damaging it, as floodwater rises on the exterior of the structure. Also, once inside the structure, floodwater could enter the crawlspace from above, turning the sealed crawlspace into a reservoir—not a good thing to do.

Wood framing (D). Testing and observation have shown that moisture levels in studs and floor framing will in most cases return to preflood levels and create no long-term problems. Autopsies of the tests showed no visible evidence of mold growth on the wood framing. Wood framing resists flood damage as long as the wall or floor system of which it forms a part will permit it to dry to normal levels (see photo).

Sheathing (E). In tests and observations, plywood sheathing stood up well, with water staining the only long-term evidence of flood exposure. However, when covered with plywood siding, the plywood sheathing dried quite slowly. After flood exposure, the slow drying of plywood sheathing might create long-term problems. It is preferable to use flood-damage-resistant lap siding with plywood sheathing, because it is more likely to permit adequate drying. Water-resistant, fiber-reinforced gypsum sheathing (for example, the Fiberock Weather Resistant (WR) Sheathing by USG that we used in our testing) main-

tained its integrity and mechanical properties. This sheathing dried to preflood levels with no visible evidence of mold growth on either the inside or the outside surface. Other WR gypsum sheathing products are likely to perform in a similar manner, although we did not test them.

Oriented strand board (OSB) sheathing swelled when subjected to flooding. We did not evaluate the impact of this dimensional change on the nailing and structural integrity of the sheathing. If flooding did not adversely affect these two factors, OSB sheathing could be an acceptable choice, because siding covers it, so its postflood appearance is unimportant.

Siding (F).

Tests showed that newly installed and painted plywood and hardboard lap siding would withstand damage but remained discolored after washing. Restoration to preflood conditions would require stain sealing and repainting. Older, weathered plywood and hardboard siding is projected to have much poorer restorability. These materials are not recommended in flood-prone areas.

Both vinyl and fiber cement sidings can withstand flooding better than hardboard lap siding and plywood siding. Vinyl and fiber cement sidings can usually be restored to preflood conditions by simply washing the portion below flood level. Older vinyl siding and painted fiber cement siding with an oxidized surface may also have to be cleaned above the flood level in order to maintain a consistent appearance. Vinyl and fiber cement siding are recommended for homes in flood-prone areas.

There was no visible evidence of mold growth from flood exposure on either the inside or the outside surface of any of the siding materials tested.

Sawn wood corner boards and trim cracked and warped after flood exposure. Replacing these items with more durable trim materials, such as plastic or wood/plastic composites, is likely to be more cost-effective than restoring them.

Wall and floor insulation (G).

Fibrous insulation, such as fiberglass or rock wool batts, contributes to higher moisture levels in the exterior wall cavities and below the floor, which can keep the adjacent walls and floor materials wetter longer. This in turn could cause



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Inspection of these exterior, insulated wall cavities revealed no damage to the studs and plywood sheathing. The fiberglass insulation was also undamaged. However, moisture levels of adjacent materials were elevated. The back side of the gypsum wallboard showed no evidence of mold growth.

long-term damage to subflooring, flooring, wall framing, and gypsum wallboard. The paper facing on batts can also slow the drying process, because the facing acts as a partial vapor barrier. We have not tested cellulose insulation under flood conditions. However, given the nature of the material, we can envision several problems. First, cellulose insulation will retain significant moisture and will probably be slow to dry. Second, the insulation may consolidate after being saturated by floodwater, leaving it full of gaps. Third, because cellulose insulation is made from the same material as drywall paper facings, which supported mold growth, it may also support mold growth within the wall cavity. Fiberglass and spray polyurethane foam (SPUF) insulations did not support mold growth. We recommend that fibrous insulation that has been subjected to floodwater be removed.

SPUF insulation tested in the wall cavities enabled the wallboard and wood studs in the exterior walls to dry at the same rate as in the interior walls with empty cavities. SPUF absorbs water very slowly and was undamaged by flooding. SPUF did not retain moisture and does not adversely effect the materials around it. There was no visible evidence of mold growth on the SPUF insulation. SPUF is recommended where insulation is expected to be subjected to flooding (see photo).



SPUF insulation performed well during Tuskegee tests. A slight scum line is the only visible evidence of flood exposure.

Interior gypsum wallboard (H).

When conventional paper-faced gypsum wallboard was tested with fiberglass batt insulation on exterior walls, it lost strength and remained wetter longer than gypsum wallboard on interior walls. The gypsum wallboard on interior partition walls dried out and maintained its strength. If gypsum wallboard is not contaminated and is able to dry completely, it can be restored to pre-flood condition with only cosmetic repairs. Although it supported mold growth on the exposed painted surface, it could be cleaned, sanitized, and restored. Visible evidence of mold growth from flood exposure on the unexposed surface of gypsum wallboard was not seen during testing and was seen only once in New Orleans.

Fiber-reinforced gypsum interior wallboard (ASTM C-1278)—a non-water-resistant product by USG called Fiberock, for example—retained most of its initial flexural strength and dried out during the drying period. Like

paper-faced gypsum board, this material supported mold growth, but it too could be cleaned, sanitized, and restored. We found no visible evidence of mold growth on the unexposed surface of this wallboard.

Water-resistant fiber-reinforced gypsum wall panels, such as USG's Aquatough, were tested; these panels maintained most of their initial strength and dried out. They did not support mold growth on either surface and were easily cleaned and restored. We assume that some other WR gypsum wall panels will perform in a similar manner, although we did not test them.

We tested water-resistant gypsum greenboard and found that it dried much more slowly than any other wallboard tested. This material is intended for use under intermittent conditions of moisture exposure, such as

showers. It is not intended for use under prolonged exposure to water, as occurs in a flood. Eventually the material dried and returned to its pre-flood condition. However, "water resistant" in this case did not mean "flood damage resistant." We do not consider greenboard to be a flood-damage-resistant option; the lengthy delay in drying out would have delayed restoration of the rest of the house and extended the time before the homeowner could move back in.

Wall finishes (I). Both latex flat paint and latex semigloss enamel paint peeled, blistered, and stained after water exposure. Mold grew on both types of paint. High- and low-permeability paints were tested. Both had to be sanded and repainted to restore the walls.

We also compared water-based flat latex and oil-based flat enamel paint. The water-based latex flaked and blistered. The oil-based flat enamel performed better than any other paint that we tested. It flaked and blistered very little

and was by far the easiest to restore. While oil-based flat enamel paint was the most resistant to flood damage, we did not investigate its effect on the drying of adjacent materials and systems.

Testing showed that standard drywall compound and paper joint tape perform very poorly under flood conditions. Most of the joints below the flood level failed completely. Quick-setting joint compound and fiberglass tape were substituted in some tests, with greatly improved results. When used with the water-resistant gypsum wallboard and oil-based paint, these joint materials required no repair after flooding.

Vinyl wall covering blistered, peeled, and came unglued after flooding. It damaged the surface of the gypsum board, and it usually inhibited drying of the wall system to the interior. Ceramic tile performed well under flood conditions and showed no long-term deterioration. However, it too inhibited drying of the wall system to the interior.

Floors (J). A sealed concrete floor slab will probably remain undamaged, unless the floodwater contains contaminants such as fuel oil or sewage. In that case, special cleanup procedures may be required. To assess the likelihood of contamination, the first question to answer is, Are contaminants normally found in or near the flooded home? If fuel oil is not used in the home, it probably won't be found in the floodwater. Sewage is potentially a problem in any home that doesn't have a sewer line backflow valve. Appearance and smell are good indicators of sewage contamination; however, they are not infallible.

T&G plywood subflooring and wood framing retain very high moisture content after a flood when unfaced fiberglass batt insulation is installed under the subfloor. Faced fiberglass insulation would probably slow the drying process further. The subflooring and framing could be subject to long-term moisture problems. With no floor insulation, the subflooring will typically return to pre-flood moisture levels fairly quickly. We tested SPUF insulation under the subfloor and found that it effectively blocked downward drying into the crawlspace. This could be a problem if an impermeable finish flooring, such as sheet vinyl, were left in place after a flood.

Although we did not test OSB subflooring, we found that when it was

used as sheathing in the walls, it swelled when it was subjected to flooding and retained an irregular surface when it dried. This characteristic would make OSB unsuitable for most subflooring applications, because the swelling and irregular surface would telescope through the finish flooring.

Floor finishes (K). Ceramic tile and quarry tile on both concrete and plywood subflooring performed well under flooding conditions and required only cleaning to be restored. However, a wood subflooring system used with these materials must have the potential to dry thoroughly, usually into the basement or the crawlspace. Floor insulation can impede this drying.

All tested carpeting—even water-resistant carpeting—and padding became dirty and smelly after flooding. It also retained large amounts of moisture, which can slow the overall drying rate throughout the house. Even if the carpet and padding is able to withstand the flood, it should be removed for cleaning and drying, and to promote drying within the home.

A simulated-wood floor—composite wood fiber and plastic—warped and had open joints larger than $\frac{1}{16}$ inch when left in place after the flood. When the simulated-wood floor panels were removed, washed, and stacked to dry after flooding, this flooring had much less warping and shrinkage, but the process of removal damaged some of the pieces.

Both glued-in-place and floating vinyl flooring on padding were tested. Both had bubbles of water trapped beneath. The padding under the floating flooring was saturated. Both systems were removed to promote drying of the subflooring. If the floating flooring can be removed without damaging it, it might be reused. Whether or not this would be cost-effective would depend on the age, condition, and value of the flooring.

Exterior doors (L). Exterior wood panel doors and exterior prehung metal-clad doors in wooden frames were stained slightly but could be washed and restored. The wood panel door was installed new and had received several coats of urethane varnish just prior to testing. We do not know whether an older door would perform as well as this one did. Fiberglass doors and foam-filled metal doors were reused



Homeowners in New Orleans discarded many appliances because they were too old or too difficult to clean.

in multiple tests, and were easily restored each time to preflood condition. After flooding, extra effort was needed to open several of the doors, because the wooden door frames had swollen. It grew easier to open them as drying continued. The joint between the outside of the door frame and the rough opening was tested left open or filled with low-expansion foam, and both versions performed acceptably. We do not recommend the common practice of filling this joint with compressed fiberglass insulation, because the insulation can retain excessive moisture.

Interior doors (M). Four types of interior door were tested: solid-wood six-panel doors; louvered-wood bifold doors; hollow-core-wood flush-panel doors; and formed-wood-composite six-panel doors. All interior doors tested

were severely stained, and some were warped, split, and/or peeling. Considering the relatively low cost of replacement, we do not consider it economically feasible to restore such doors.

Windows (N). All of the vinyl and aluminum window frames tested were restored to preflood conditions with minimal cleaning effort. Flooding did not affect the glazing or the operation of the movable parts. No fogging of the interior of the glazing occurred. The joint between the outside of the window frame and the rough opening was tested left open or filled with low-expansion foam, and both performed acceptably. We do not recommend the practice of filling this joint with compressed fiberglass insulation, because the insulation can retain excessive moisture. Wood-framed windows were not tested. Some wood-framed windows appeared to have survived in New Orleans, but these may have swollen and become difficult

to open or shut. Other wood-framed windows in poorer condition were clearly candidates for replacement. Most of these were probably leaky and contained only single-pane glass, so replacing them would save energy.

Kitchen and bath cabinets (O). We tested a typical laminated fiberboard base cabinet to determine both how it would perform and how its performance would affect adjacent areas of the kitchen. The fiberboard elements of the cabinet readily absorbed water and swelled. This caused the laminated surface to come off. While the fiberboard did eventually dry, it was permanently disfigured and was also stained by the floodwater. Similar results were observed on solid-wood cabinets in New Orleans.

What was equally significant was the fact that the base cabinet slowed the

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drying of the adjacent wallboard. These hidden areas of wallboard did support mold growth. Some mold also grew on the uncoated parts of the base cabinet itself. If the base cabinets are not removed shortly after the flood, special provisions must be made to accelerate the drying of the adjacent walls. These normally hidden areas should be visually inspected as part of the recovery process.

Restoring Building Equipment and Appliances

The homeowner or a designated contractor must decide either to clean and refurbish or to replace equipment and appliances. Furnaces, air conditioners, and water heaters that are still fairly new are candidates for refurbishment. Older units nearing the end of their service life should be replaced. Among appliances, clothes washers and dryers with most of their service life remaining could be refurbished. Appliances dealing with food—ranges, dishwashers, and refrigerator/freezers—should be replaced regardless of age, unless the homeowner is sure that the floodwater was not contaminated, and that they can be restored to a sanitary condition. Knowing the cost of cleaning and refurbishing versus the cost of replacement will help the homeowner to make prudent replacement decisions.

Flooded water heaters with fiberglass insulation are difficult to clean and dry. Cleaning or replacing the controls and burner and then running the heater on the hottest temperature for a few days might dry out the insulation, but it could create problems if the floodwater carried contaminants. At the very least, the smell may be a problem, since some of the water in the New Orleans flooding contained sewage and decaying vegetation. We do not know anyone who has tried this approach.

Flooded equipment and appliances should be replaced if necessary, and all equipment and appliances should be installed above projected future flood levels if possible. In single-story houses, this may entail placing them on raised platforms if the projected future flood level is 2 ft or less. Moving the furnace, air conditioner, and water heater to the attic will keep them above higher projected floods. In two-story houses, they

could be installed on the second floor, and their respective distribution systems could be installed in the interstitial space between floors.

More specific recommendations follow.

Furnace/air conditioner (P). If the furnace and air conditioner cannot be cleaned and refurbished, or if they are near the end of their service life, we recommend that they be replaced with appropriate new units (consider Energy Star) that are located on a raised platform, on the second floor, or in the attic to keep them above projected future flood levels. Drip pans with a visible overflow drain can be installed when the air conditioner is moved to the second floor or attic. This will reduce the potential for water damage from a plugged condensate drain line.

It is important that the replacement unit and its duct system be properly sized. Some energy upgrading will probably be done during reconstruction. The system should be downsized to reflect the smaller loads and to achieve the best dehumidification cycles. This is particularly important in the humid South.

When moving the refurbished or replacement unit to the attic, the contractor should be careful to provide good access for maintenance, repair, and ultimate replacement. Filters should be located where they are easy to remember and replace. Typically this would be on an inhabited floor at the return air grille.

Ductwork (Q). Ductwork located below the first floor, in both slab and crawlspace houses, will probably have been damaged by the flood. At the minimum, the duct system will be hard to clean. Replacement ducts should be relocated to the attic, above a dropped ceiling in a central hallway, or in the interstitial space between the first and second floors for two-story houses if possible. Metal ducts that remain within the crawlspace, especially return air ducts, should be sealed and reinsulated after they have been cleaned, sanitized, disinfected, and repaired, as necessary. This will prevent excess moisture and mold from being drawn from the crawlspace into the house.

Water heater (R). If a water heater cannot be cleaned and refurbished, or if

it is near the end of its service life, it should be replaced with an appropriate new unit located on a raised platform, on the second floor, or in the attic to keep it above projected future flood levels. Instantaneous, or tankless, water heaters may be a desirable option when the water heater is relocated to the attic. These require less space and weigh less than conventional water heaters. Spillage pans with a visible overflow drain must be installed when the water heater (especially a tank-type unit) is moved to the second floor or attic. This will reduce the potential for water damage from a leaky water heater that has reached the end of its service life. When moving the replacement unit to the attic, the contractor should be careful to provide good access for maintenance, repair, and ultimate replacement (see photo, p. 26).

Undamaged hot water distribution systems in crawlspaces or below floor slabs can be reused. However, if damage has occurred, the replacement system should be located in the attic or in the interstitial space between the first and second floors for two-story houses. Usually this will also reduce energy loss from the distribution system, and the waiting time for hot water to arrive at the fixture.

Associated attic modifications (S). When the furnace, air conditioner, and water heater are relocated to the attic, modifications to the attic must be considered. The attic floor may need to be strengthened to carry the added load of this equipment. In cold climates, where the attic temperature will fall to freezing and below, the equipment and water piping must be insulated and freeze protected if the attic is ventilated.

Another approach that is useful in all climates is to bring the attic into the conditioned space. This involves closing off and sealing all vents and placing insulation on the gable ends and under the roof deck. This eliminates the need to freeze protect pipes and equipment and reduces the ill effects of leaky ductwork. Keep in mind, though, that if combustion equipment, such as a gas furnace, is placed in an unvented attic, provision must be made to supply combustion air.

An unvented attic must be insulated correctly, to prevent moisture problems. Some type of fibrous insulation—fiberglass or rock wool, for example—should

be installed on the underside of the roof deck. Besides costing less than SPUF, fibrous insulation will let the deck breathe and dry if rainwater ever penetrates the roof membrane.

Laundry appliances. If laundry appliances cannot be cleaned and refurbished or if they are near the end of their service life, they should be replaced with appropriate new units (consider Energy Star), either installed on a raised platform or relocated to the second floor, if there is one. The more energy- and water-efficient horizontal-axis, front-loading washers and front-loading dryers would work well on a raised platform. Another option is to stack the dryer above the washer, as is done in apartments with limited laundry space. This should protect at least the upper unit from flood damage.

Kitchen appliances. Because kitchen appliances usually cannot be relocated, there is little that can be done to reduce their exposure to future floods. Replacement or refurbished units should be installed in such a way that they can be easily removed to allow the adjacent areas to dry in the event of a future flood.

Electrical system (T). The main breaker panel should be relocated (if possible) to a point at least 1 foot above the highest projected future flood level, while maintaining reasonable access in case of a tripped circuit. If the electrical distribution system has been damaged and requires replacement, the new system should be located as high as possible above the projected future flood level. Placing feeding switches and receptacles above the projected flood level will reduce the need to replace them. While testing at Tuskegee University has shown that modern wiring with plastic insulation was not harmed by submersion, most current electrical codes require that all wiring that has been submerged be replaced. Receptacles and switches can corrode or become dirty if they have been submerged. However, it costs relatively little to replace switches and recepta-

cles, so it is not worth the effort to raise them above the projected flood level should they become damaged.

Sanitary sewer (U). Install a backflow valve on the sanitary sewer line to reduce the chance of sewage backing up into the house if there is another flood.



A flood-conscious New Orleans homeowner located the water heaters in the attic in an easily accessible location. They were completely undamaged by the flooding.

The Costs and Benefits for Homeowners

Using flood-damage-resistant restoration techniques will cost somewhat more initially than replacement in kind. Our rough estimates indicate that the added cost will be about 10%–15% of the total cost of restoration. Offsetting this added initial cost are the lower energy bills associated with the improved insulation and air sealing of the building envelope, the smaller replacement heating and cooling equipment, and the improvements to the duct system resulting from reconstruction. The payback from these savings will offset the higher initial cost.

The overall benefits include reduced vulnerability to future flood damage and disruption, and potentially lower home flood insurance rates. Restoration from any future floods will be much less costly, and the time to reoccupy the flooded home much shorter, if these wet floodproofing recommendations are implemented. Finally, the homeowner will enjoy not only lower energy bills, but also increased comfort in the newly restored home.

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For more information:

To obtain a copy of the book, *Repairing Your Flooded Home*, published by the American Red Cross, go to www.redcross.org/static/file_cont333_lang0_150.pdf.

For the full technical report that forms the basis of this article, *Field Testing of Energy-Efficient Flood-Damage-Resistant Envelopes*, go to www.ornl.gov/sci/res_buildings/NaturalDisaster.htm.

For information on FEMA flood-related policies, go to www.fema.gov/nfip.